AI SPEECH INTELLIGENCE

Research Roadmap for Multilingual Household Interviews

Focus: speech-to-text (ASR), multilingual/code-switched speech, speaker diarization, audio quality, and evaluation.

Research objective

Determine which practical speech pipeline can turn consented household-interview recordings into accurate, speaker-labelled transcripts while balancing quality, processing time, cost, privacy, and real-world recording conditions.

This is a research plan, not a claim that any provider is automatically best. Model performance must be measured on recordings representative of the intended use.

A step-by-step guide for the first research phase.

1. Understand the System

An end-to-end speech intelligence workflow has several components. Keep them separate during research so you can identify which stage causes an error.

Component	Purpose	Example
Audio recording	Capture interview audio after informed consent; preserve the original securely.	Field worker records a conversation.
ASR / speech-to-text	Convert spoken audio into text.	Telugu speech becomes a Telugu transcript.
Language handling	Handle language changes within a sentence or conversation.	Telugu mixed with English or Hindi.
Speaker diarization	Estimate who spoke when, using anonymous labels.	speaker_1 and speaker_2 with timestamps.
Transcript review	Let a reviewer correct uncertain words and speaker labels.	Correct a place name or number.

Important: diarization estimates different speakers; it does not establish their real-world identities. Survey consent and permission to record audio should be treated separately.

2. Research Area A — Speech-to-Text (ASR)

Core question: Which system produces the most accurate transcript for the languages, dialects, and recording conditions in your household interviews?

Candidates to evaluate

Candidate	What to investigate
Sarvam AI	Relevant Indian languages, code-switching, noisy audio, API limits, data handling, cost, and latency.
Whisper / hosted implementations	Multilingual performance, language detection, timestamps, deployment options, and operational cost.
AI4Bharat models and datasets	Model availability, language coverage, licenses, evaluation data, compute needs, and deployment requirements.
Google Gemini or another audio-capable model	Verify the exact model/API's audio workflow, languages, output format, limits, privacy terms, and pricing.

Use official documentation to verify current model names, formats, limits, pricing, and data-processing terms before deciding.

Suggested experiment conditions

• Telugu-only conversation.

• Telugu-English code-switching.

• Telugu-Hindi-English mixed speech.

• Different Telugu accents or dialects, where appropriate and consented.

• Background noise, low volume, fast speech, pauses, and overlapping speech.

• Names, locations, dates, ages, household counts, and monetary amounts.

3. Research Area B — Speaker Diarization

Diarization divides audio into speaker-labelled time segments. It can help separate interviewer and respondent, but labels can be wrong when people interrupt, overlap, or have similar voices.

Illustrative output

```
{ "segments": [ {"speaker_id": "speaker_1", "start": 0.0, "end": 3.2, "text": "How many people live in your household?"}, {"speaker_id": "speaker_2", "start": 3.3, "end": 5.1, "text": "There are five people."} ] }
```

Research questions

• How well does the model detect speaker changes?

• What happens when two people speak at the same time?

• Does diarization quality change for Telugu or mixed-language speech?

• How often are interviewer and respondent turns assigned incorrectly?

• Can timestamps make manual review easier?

4. Research Area C — Multilingual and Code-Switched Speech

Speakers may switch languages within one sentence. Evaluate whether the system preserves the words actually spoken, represents scripts consistently, and avoids silently translating or inventing content.

Example utterance: "■■ ■■■■■■ five members ■■■■■■■, ■■■■■■ children school ■■ ■■■■■■■■."

• Language switching within a sentence or turn.

• Telugu script versus English words and numerals.

• Names of people, villages, streets, and local institutions.

• Numbers, ages, dates, family counts, and amounts.

• Dialect differences and regional pronunciation.

• Background noise and overlapping speech.

5. Research Area D — Audio Quality and Recording Design

Test with the phones and environments field workers will actually use. Studio recordings alone do not demonstrate real-world performance.

• Microphone distance and device differences.

• Road, fan, television, and crowd noise.

• Quiet speech, clipping, and low volume.

• Long pauses, interruptions, and overlapping speakers.

• Audio formats, compression, file size, and upload failures.

• Interrupted recording, app restart, offline storage, and later upload.

6. Evaluation Metrics

Compare each candidate against a human-verified reference transcript. Report results by language and recording condition, not only as one overall score.

Metric	What it measures	Caution
WER	Word substitutions, deletions, and insertions.	Tokenization and script conventions affect comparisons.
CER	Character-level edit errors.	Low CER does not guarantee correct meaning.
Named-entity accuracy	Correct names and place names.	Requires consistent annotation rules.
Number accuracy	Correct ages, counts, dates, and amounts.	Report critical-number errors separately.
Diarization error	Errors assigning speech time to speaker labels.	Define how overlap and unknown speakers are scored.
Latency and cost	Processing time and spend per audio minute.	Include retries, storage, and review costs.

WER/CER should use a documented text-normalization policy. Keep the original transcript and normalization rules so results are reproducible.

7. First Practical Research Project

Project title: Comparative Evaluation of Multilingual Speech Recognition for Telugu Household Interviews

Suggested starting protocol

1. Define the scope. Choose the languages, dialects, interview style, devices, and deployment constraints to support.

2. Build a consented test set. Start with 50–100 short recordings if feasible. Include multiple speakers and realistic conditions. Never record people without informed permission.

3. Create reference transcripts. Have proficient annotators transcribe recordings, review uncertain sections, and agree how to represent mixed-language speech, numbers, and names.

4. Select candidates. Choose two or three systems. Check official documentation, supported languages, API limits, cost, licenses, and privacy terms.

5. Run identical tests. Submit the same audio to each candidate using comparable settings. Save model/version, parameters, output, latency, and cost.

6. Score and categorize errors. Calculate WER/CER and separately inspect names, numbers, code-switching, noise, and speaker attribution.

7. Review failures. Manually inspect errors and note whether they came from recording quality, ASR, diarization, or annotation disagreement.

8. Decide what to build next. Choose based on measured quality, privacy, reliability, cost, and deployment needs.

The suggested 50–100 recordings are a practical pilot size, not a universal statistical requirement. Expand the test set before making broad claims across many dialects, devices, or environments.

Suggested experiment tracking table

Recording ID	Condition	System/version	WER/CER	Name/number errors	Latency/cost	Notes
R001	Telugu only	Candidate A	—	—	—	—
R002	Telugu + English	Candidate A	—	—	—	—
R003	Mixed + noise	Candidate B	—	—	—	—

8. Recommended Research Sequence

Phase	Focus	Deliverable
1	ASR for Telugu and code-switching	Baseline results and error taxonomy
2	Speaker diarization	Speaker-labelled segments and error analysis
3	Audio conditions and mobile recording	Recording guidance and failure-handling requirements
4	Privacy, retention, and deployment	Recording, storage, access, and deletion workflow
5	Pipeline integration	Reviewed transcript service ready for survey extraction

9. Research Resources

Starting points; confirm current availability, paper details, licenses, model versions, and data terms from the official source.

- **AI4Bharat:** ai4bharat.iitm.ac.in
- **IndicVoices dataset:** Hugging Face dataset
- **IndicContextEval:** GitHub repository
- **Indic DiarBench paper:** arXiv paper
- **Indic DiarBench dataset:** Hugging Face dataset
- **Sarvam AI:** Official website

- **Hugging Face:** Models and datasets
- **Google Scholar:** Research search
- **arXiv:** Research preprints

10. Research Checklist

- ■ Target languages and recording conditions are defined.
- ■ Recordings were collected with informed consent and secure handling.
- ■ Reference transcripts have been reviewed.
- ■ The same audio was tested on each candidate system.
- ■ Exact model versions and settings were recorded.
- ■ WER/CER and name, number, and speaker-label errors were checked.
- ■ Cost and latency were measured for the intended workflow.
- ■ Privacy, retention, deletion, and third-party processing risks were documented.
- ■ The next step was chosen based on evidence rather than assumptions.

Final recommendation

Start with a controlled ASR comparison for Telugu and code-switched interviews. After identifying transcription errors, evaluate speaker diarization. Then integrate the outputs into the survey workflow with human review for uncertain or consequential details.